Polynomial Factorisation

Exam code: 9MA0

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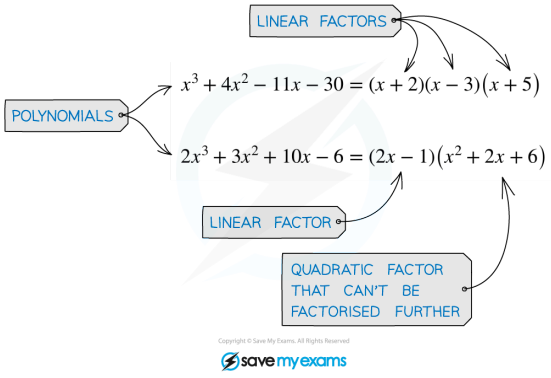
Polynomial factorisation

What is polynomial factorisation?

- Factorising a polynomial combines the **factor theorem** with the method of **polynomial division**
- The goal is to break down a polynomial as far as possible into a product of linear factors



How do I factorise a polynomial?

- At A level you will usually be asked to factorise a **cubic** – i.e. a polynomial where the highest power of ***x*** is 3

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$3 - x^2 - 4x^3$$

$$x^3 + 2x$$

$$2x^3$$

- To factorise a cubic polynomial **f(*x*)** follow the following steps:

> ? Fully factorise $x^3 + 4x^2 - 11x - 30$.

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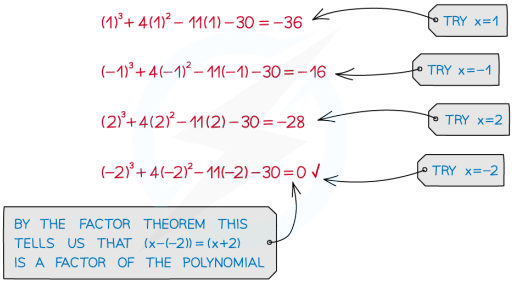
- Step 1. Find a value ***p*** that makes **f(*p*) =** 0



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- Step 2. Use polynomial division to divide **f(*x*)** by **(*x* – *p*)**

$$
\begin{array}{r|l}
 & x^2 + 2x - 15 \\
\hline
x+2 & x^3 + 4x^2 - 11x - 30 \\
 & -(x^3 + 2x^2) \\
\hline
 & 2x^2 - 11x - 30 \\
 & -(2x^2 + 4x) \\
\hline
 & -15x - 30 \\
 & -(-15x - 30) \\
\hline
 & 0
\end{array}
$$

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- Step 3. Use the result of your division to write

$$\mathbf{f(x) = (x - p)\,(ax^2 + bx + c)}$$

SO $x^3 + 4x^2 - 11x - 30 = (x+2)(x^2 + 2x - 15)$

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- Step 4. If the quadratic **($ax^2 + bx + c$)** is factorisable, factorise it and write **f(*x*)** as a product of three linear factors (if the quadratic is not factorisable, then your result from Step 3 is the final factorisation)

Examiner Tips and Tricks

- The method outlined above can be logically extended to factorise a polynomial of any degree.

Worked Example

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